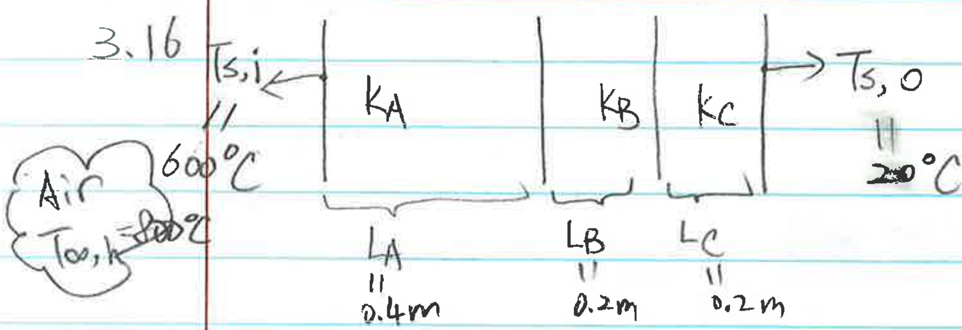
CHE 312
HW 4.

90/100

Fu Wu

3.16

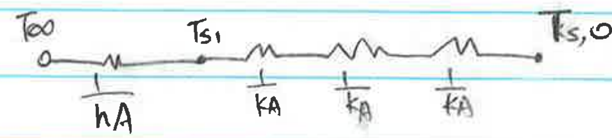


$k_A = 25 \, \mathrm{W/m \cdot K}$

$k_C = 60 \, \mathrm{W/m \cdot K}$

$h = 25 \, \mathrm{W/m^2 \cdot K}$

$k_B = ?$



$$R_{cond,A} = \frac{L_A}{k_A \cdot A} = \frac{1}{A}\left(\frac{L_A}{k_A}\right) = \frac{1}{A}\left(\frac{0.4\,\mathrm{m}}{25\,\mathrm{W/m \cdot K}}\right)$$

area

$$R_{cond,B} = \frac{L_B}{k_B \cdot A}$$

$$R_{cond,C} = \frac{L_C}{k_C \cdot A}$$

(25/25)

$$R_{cond,tot} = \frac{L_A}{k_A \cdot A} + \frac{L_B}{k_B \cdot A} + \frac{L_C}{k_C \cdot A}$$

$$= \frac{1}{A}\left(\frac{L_A}{k_A} + \frac{L_B}{k_B} + \frac{L_C}{k_C}\right)$$

$$= \frac{1}{A}\left(\frac{0.4\,\mathrm{m}}{25\,\mathrm{W/m \cdot K}} + \frac{0.2\,\mathrm{m}}{k_B} + \frac{0.2\,\mathrm{m}}{60\,\mathrm{W/m \cdot K}}\right) = \frac{1}{A}\left(0.016\,\frac{\mathrm{m^2 K}}{\mathrm{W}} + \frac{0.2\,\mathrm{m}}{k_B} + \frac{1\,\mathrm{m^2 k}}{300\,\mathrm{W}}$$

✓

$$q = \frac{T_{s,i} - T_{s,o}}{R_{cond\,tot}}$$

$$\left[q = \frac{(600-20)\,\mathrm{K}}{\frac{1}{A}\left(0.016 + \frac{0.2}{k_B} + \frac{1}{300}\right)} \right] \times \frac{1}{A} \rightarrow \frac{q}{A} = q'' = \frac{580\,\mathrm{K}}{\left(0.016 + \frac{0.2}{k_B} + \frac{1}{300}\right)}$$

$r = \frac{D}{2} = 1mm = 1\times10^{-3} m$

3.43)

$2mm \quad R = 0.01 \Omega/m \quad I = 20A$

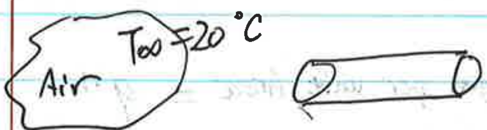
a) $V = IR$

$q^{①} = P = I \cdot V = I^2 \cdot R = (20A)^2 \cdot (0.01 \Omega/m) = 4 J/s \cdot m = 4 \frac{W}{m}$

$q^{②} = q^{①} \cdot \frac{1}{A} = 4 \frac{W}{m} \cdot \cancel{\ldots} \frac{1}{\pi \cdot (1\times10^{-3} m)^2}$

$= 1273239.545 \frac{W}{m^3}$

b)



$\varepsilon = 0.3$

$h = C[(T - T_\infty)/D]^{\frac{1}{4}}$

$C = 1.25 \frac{W}{m^{7/4} \cdot K^{5/4}}$

$q^{①} \div (\pi \cdot D) = q''_{conv} + q''_{rad}$

(heat gen pre unit surface Area) $[\frac{W}{m^2}]$

$h(T_s - T_\infty) + \varepsilon\sigma(T_s^4 - T_\infty^4)$

°C → K +273.15

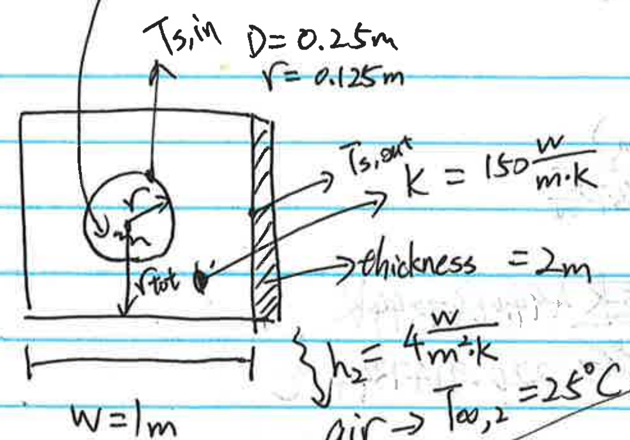
(18/25)

$4 \frac{W}{m} \div (\pi \times 2\times10^{-3} m) = 1.25 \left(\frac{T_s - 20}{D}\right)^{\frac{1}{4}} (T_s - 20) + 0.3 \times 5.67\times10^{-8} \frac{W}{m^2K^4} \times (T_s^4 - 20^4)$

$636.62 \frac{W}{m^2} = 1.25 \left(\frac{(T_s - 293.15)K}{2\times10^{-3} m}\right)^{\frac{1}{4}} (T_s - 293.15)K + 1.701\times10^{-8}(T_s^4 - 293.15^4)$

$T_s = 331.1521 K = 58.0021 °C$

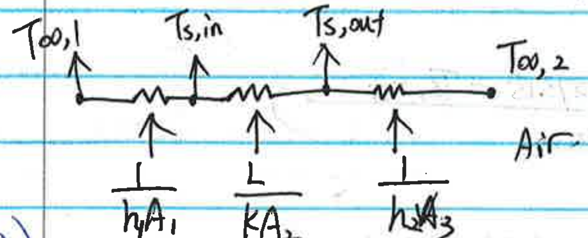
4.28



inside the hole:

$T_{\infty,1} = 300°C$

$h_1 = 50 \frac{W}{m^2 \cdot K}$



I'm assuming the distance from center to the surface of the solid block is constant r_{tot} (shows in the graph)

so L in the eq $= r_{tot} - r$

$= \frac{W}{2} - r$

$= \frac{1}{2}m - 0.125m$

$= 0.375m$

(22/25)

A_1 in the eq $= \pi D \cdot$ thickness

$= \pi \cdot 0.25m \cdot 2m$

$= 1.5708\ m^2$

~~assume $A_1 \approx A_2$~~

$A_3 = 1 \times 2 \times 4 = 8m^2$

$A_2 = \cancel{\frac{2\pi k}{\ln(W/D)}}$

$= \frac{2\pi L}{\ln(1.08W/D)}$

$= 8.58m^2$

$$q = \frac{T_{\infty,1} - T_{\infty,2}}{\frac{1}{50\frac{W}{m^2 \cdot K} \cdot 1.5708m^2} + \frac{0.375m}{150\frac{W}{m \cdot K} \cdot 8.58m^2} + \frac{1}{4\frac{W}{m^2 \cdot K} \cdot \cancel{1.5708m^2}\ 8}}$$

$$q = \frac{[(300+273.15) - (25+273.15)]K}{\cancel{0.1734785 \frac{K}{W}}\ \cancel{0.045574 \frac{K}{W}}\ 0.044274\frac{K}{W}}$$

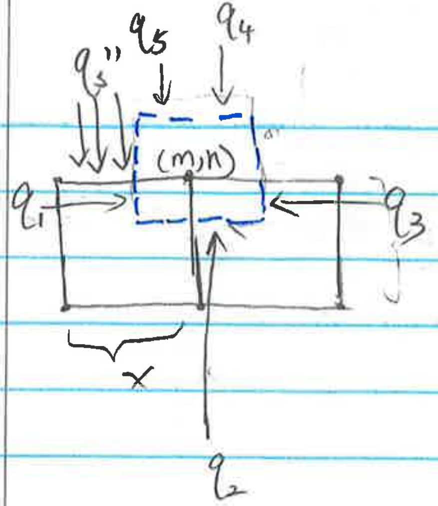
$\cancel{= \frac{275}{0.1734785}} \cancel{= 1585.2085W} = \frac{275K}{0.044274\frac{K}{W}} = \cancel{6034.154005W}$

$q = 6211.35675W$

q'' will be constant ~~through the~~

$q'' = \frac{q}{A_1} = \cancel{\frac{6034.154005W}{1.5708m^2}} \cancel{= 3841.452766\frac{W}{m^2}}\ \frac{6211.35675W}{1.5708m^2} = 3954.2\ \frac{W}{m^2}$

4.35)



$$\sum_{i=1}^{5} q_i + \cancelto{0}{\dot{q}}(\Delta x \cdot \Delta y \cdot 1) = 0$$

$$q_1 = k\frac{(T_{m-1,n} - T_{m,n})}{\Delta x}\left(\frac{\Delta y}{2}\cdot 1\right)$$

$$q_2 = k\left(\frac{T_{m,n-1} - T_{m,n}}{\Delta y}\right)(\Delta x \cdot 1)$$

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$$q_3 = k\left(\frac{T_{m+1,n} - T_{m,n}}{\Delta x}\right)\left(\frac{\Delta y}{2}\cdot 1\right)$$

$$q_4 = h(T_\infty - T_{m,n})\left(\frac{\Delta x}{2}\cdot 1\right)$$

$$q_5 = q_s'' \cdot \left(\frac{\Delta x}{2}\cdot 1\right)$$

set $\Delta y = \Delta x$

$$\Rightarrow \Big[k(T_{m-1,n} - T_{m,n})\cdot\tfrac{1}{2} + k(T_{m,n+1} - T_{m,n}) + k(T_{m+1,n} - T_{m,n})\cdot\tfrac{1}{2} + h(T_\infty - T_{m,n})\Delta x\cdot\tfrac{1}{2} + q_s''\left(\tfrac{\Delta x}{2}\right) = 0 \Big] \times 2$$

(answer is in the back)